

A Reusable Agentic Platform for Auditable, Sentiment-Aware Dynamic Asset Allocation

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Project Proposal

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1. Motivation

Banks and asset managers are turning to AI to help analysts, portfolio managers, and risk teams get through information-heavy work. In these settings, though, the system has to stay under human control: it needs to be auditable, explainable, and robust, and it has to leave the final call to a person. That is also where regulators have landed. The recent supervisory debate on AI in finance is about governance, documentation, risk management, and human oversight, not about handing decisions to a model [1], [2], [3].

The split between stocks and bonds is one of the biggest decisions an asset manager makes. That single high-level allocation drives most of a portfolio's risk and return, and many firms revisit it on a regular schedule (a dynamic, or tactical, allocation process), using quantitative signals (valuations, momentum, macro and rates data, risk models) together with a read on the prevailing market narrative. Quantitative models handle the measurable side well. What they rarely capture is how a shift in narrative, news sentiment, or market attention should feed into a tactical tilt. Generic LLM assistants have the opposite weakness: they can summarize a narrative, but they offer no deterministic analytics, no traceability, no link to a firm's own models, and little workflow control.

This project is run with an asset-management partner that wants a system to support exactly this decision. We propose a **reusable agentic platform for finance** and test it on one concrete use case: **auditable, sentiment-aware recommendations for a portfolio's bond/stock split**, written up as an investment-committee report. Where possible, the platform incorporates some of the tools and models the firm already uses; where those cannot be shared or integrated directly, it aims to reproduce similar behaviour through stand-ins with the same interface. On top of that it adds a sentiment-simulation layer and orchestrates the pipeline. It does not try to rebuild the firm's quantitative stack.

The system does not trade, move the allocation, give personalized advice, or stand in for the investment committee. It supports the people who do those things: gathering evidence, building and simulating sentiment scenarios, working out what a candidate allocation does to the portfolio, checking its own claims, and producing a recommendation memo with an audit trail behind it.

2. Applied Research Question

How can a reusable LangGraph-based agentic platform support an investment committee in producing auditable, sentiment-aware recommendations for a portfolio's dynamic bond/stock allocation, by combining the firm's existing

quantitative models, source-grounded market narratives, multi-agent sentiment simulation via a dedicated agentic engine, and deterministic portfolio risk analytics?

This is an applied question, not a purely academic one. The point is to build and evaluate a working prototype, and to see whether agentic AI can be useful, controllable, and inspectable in a realistic allocation workflow the partner actually cares about.

3. Project Objectives

The project has five objectives.

A reusable platform. We design an agentic platform for financial workflows that handles orchestration, state, tool use, memory, validation, and human-in-the-loop control, and that can be extended beyond the allocation case. LangGraph fits well here, since it is built for long-running, stateful agent workflows with persistence, durable execution, tracing, and human oversight [4].

One concrete workflow. On top of the platform we build a dynamic bond/stock allocation workflow. Given a sandbox portfolio, a mandate, the firm’s signals, and a market context, it retrieves the relevant information, builds and simulates sentiment scenarios, computes the risk and return of candidate equity/bond mixes, and writes a memo recommending whether and how far to shift the split.

Integration and simulation. We look at how the platform can plug into the partner’s existing tools and models, and how a sentiment-simulation component fits alongside them. The aim is to incorporate some of the firm’s models, signals, and feeds as callable tools where that is feasible, and otherwise to provide similar behaviour through stand-ins rather than reimplementing the firm’s stack. The exact engine is left open: the sentiment-simulation component is provided by an external multi-agent simulation engine such as OASIS, a multi-agent social simulation framework [5]. We use simulation cautiously: not to forecast prices or returns, but to model how sentiment spreads, how stakeholders react, and what second-order effects a narrative shift might have. We keep it an optional, isolated component, with a small internal adapter we write ourselves as the default.

Reasoning module. The platform needs a reasoning step that turns the gathered evidence into a bounded *outlook*: a view on the bond/stock tilt with explicit uncertainty that feeds the deterministic analytics. In its basic form this is a single step. If time allows, we extend it with the macro/micro decomposition from the Nexus forecasting framework [6], where a macro step reads the prevailing regime and a micro step weighs near-term catalysts before the two are combined. Either way the module is an input to the analytics, not a price or return predictor and not the recommendation on its own, and we deliberately leave out the backtest-driven calibration loop.

Evaluation. We evaluate the prototype on a post-cutoff backtest and against transparent baselines rather than a chat assistant: a static benchmark split (e.g. a fixed 60/40) and a traditional indicator-driven rule set built on classic signals such as equity valuations (P/E ratio or equity earnings yield versus bond yields), Fed rate-path expectations, and US labour-market data. The exact baseline is fixed after a short literature review and discussion with the practical partner’s experts. Following Nexus [6], evaluation runs on data after the models’ knowledge cutoff to avoid leakage, and looks at decision quality together with source grounding and auditability.

4. Proposed System

The prototype has seven main components.

1. **Agentic orchestration layer.** A LangGraph supervisor coordinates the specialized agents, holds the workflow state, logs intermediate output, and enforces the human approval points. This is the reusable core of the platform.
2. **House-model and data integration layer.** Where feasible, this incorporates some of the firm’s quantitative models, signals, and data feeds as deterministic tools the platform can call. Where a model or feed cannot be shared or integrated directly, a stand-in with the same interface provides similar behaviour. This is the piece that connects the platform to the firm’s existing tools.
3. **Research and RAG agent.** Retrieves and summarizes relevant financial news, macro and rates commentary, company filings, and market reports. Claims in the memo are linked back to their sources wherever possible.
4. **Sentiment scenario agent.** Turns retrieved information and user-defined themes into structured sentiment scenarios for the equity/bond decision: a risk-off rotation out of equities, a hawkish central-bank surprise, a growth scare, weak credit sentiment, a geopolitical escalation, or a move away from high-duration assets.
5. **Sentiment-simulation module.** Simulates how different market participants might react to an initial narrative shock. The output is a structured picture of exposure (which sectors, factors, or parts of the split are most affected) that feeds the reasoning rather than acting as a trading signal or a return forecast. It runs through an integration with an external agentic engine such as OASIS or, by default, a small internal adapter we write ourselves.
6. **Allocation analytics tools.** Deterministic Python tools that compute the risk and return of candidate bond/stock mixes: returns, volatility, drawdowns, VaR/CVaR, concentration, factor and sector exposure, benchmark-relative metrics, scenario losses, and how sensitive the split is to its inputs. Classical methods such as mean-variance optimization or risk parity [7] give a transparent reference allocator, but any optimization stays secondary to the decision-support framing.
7. **Validation, reporting, and audit layer.** A validation agent flags unsupported claims, separates fact from assumption, notes the limitations, and drafts the committee memo. The audit layer records sources, tool calls (including which house models ran), model versions, scenario and simulation assumptions, intermediate outputs, validation checks, and human approvals.

4.1. Reasoning backbone

At its core the reasoning step is a single component that produces the outlook: it reads the gathered evidence and returns a bounded view on the tilt with explicit uncertainty. It registers as one *outlook* tool that feeds the deterministic analytics and the memo. It does not predict prices or returns, and it does not produce the recommendation by itself.

As an optional extension, if time allows, we structure that step as the macro/micro split from Nexus [6]:

- A **macro-reasoning step** sets the broad regime behind the bond/stock decision (the rate cycle, the growth-versus-inflation backdrop, and the overall risk-on/risk-off posture), which gives the direction of the tilt.

- A **micro-reasoning step** weighs near-term catalysts: data releases, central-bank meetings, earnings, and the sentiment shocks the simulation module surfaces.
- A **synthesizer step** merges the two into one view and spells out how they were weighed.

Either way we leave the backtest-driven calibration loop out of scope.

Each run produces two artifacts:

1. an **investment-committee allocation memo**: whether and by how much to change the bond/stock split, with the rationale, scenario analysis, and the assumptions and limits behind it;
2. an **audit bundle**: the run configuration, data sources, tool and model calls, simulation assumptions, validation results, a claim-to-evidence matrix, and the recorded human approvals.

5. Expected Deliverables

The list is trimmed to what a working, evaluable prototype actually needs:

- the **reusable agentic platform** (LangGraph orchestration: workflow state, tool registry, validation, human-in-the-loop control, and audit logging);
- the **dynamic bond/stock allocation workflow** built on the platform, incorporating the partner’s models and data where feasible (or stand-ins with the same interface), producing the **committee memo and its audit bundle**;
- **deterministic allocation analytics tools, with tests**;
- the **reasoning module** that produces the bounded outlook, with an optional macro/micro split inspired by Nexus [6] (no calibration) if time allows;
- the **sentiment-simulation component**, either an external agentic-engine integration (e.g. OASIS) or, by default, a minimal internal adapter we write ourselves;
- the **evaluation**: a post-cutoff backtest harness, the two baselines (static benchmark and traditional indicator-based allocation), and the comparison against the platform, with component ablations and reasoning-quality results;
- a **final report** covering architecture, integration, evaluation, limitations, governance, and extendability.

6. Evaluation Plan

The evaluation is kept light. Backtests are used to sanity-check and stress the decision logic, not to claim investment performance or present a tradeable strategy.

Baselines. The platform is compared against two transparent baselines:

1. a **static benchmark allocation**, a fixed split such as 60/40, standing in for taking no active view; and
2. a **traditional allocation baseline**, a deterministic rule set that maps classic indicators (for example equity valuations, Fed rate-path expectations, and US labour-market data) to a tilt. Its exact specification follows a short literature review and discussion with the practical partner’s experts, and is frozen before any results are looked at, so the comparison stays fair.

Post-cutoff backtesting. Following Nexus [6], evaluation runs only on data after the language models’ knowledge cutoff, so a model cannot simply recall a known outcome. This is the main leakage control.

What we measure. On each window we compare the proposed tilts with the baselines on a small set of risk-adjusted and stability measures, for example realized volatility, maximum

drawdown, a Sharpe-type ratio, and the directional hit rate of the tilt. We also check that the memo is well-grounded: each material claim should trace through the audit bundle to a source, a tool output, or human input. Finally, the partner and supervisors give a short human judgment of whether the memo is committee-ready.

7. Scope and Limitations

The project does not cover live trading, allocation execution, personalized advice, or production-grade compliance, and it uses public, synthetic, or sandbox/anonymized data. The prototype is decision support for the investment committee, not an autonomous system. The scope is deliberately narrow: one platform and one evaluated workflow (the bond/stock split), though the platform is designed to extend to other finance workflows later.

On the sentiment simulation, the default path is a minimal internal adapter we write ourselves, with an external agentic-engine integration (such as OASIS) treated as an optional, isolated experiment. The engine is not a price-prediction engine; its role is to generate structured sentiment and stress scenarios that complement the deterministic analytics.

References

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